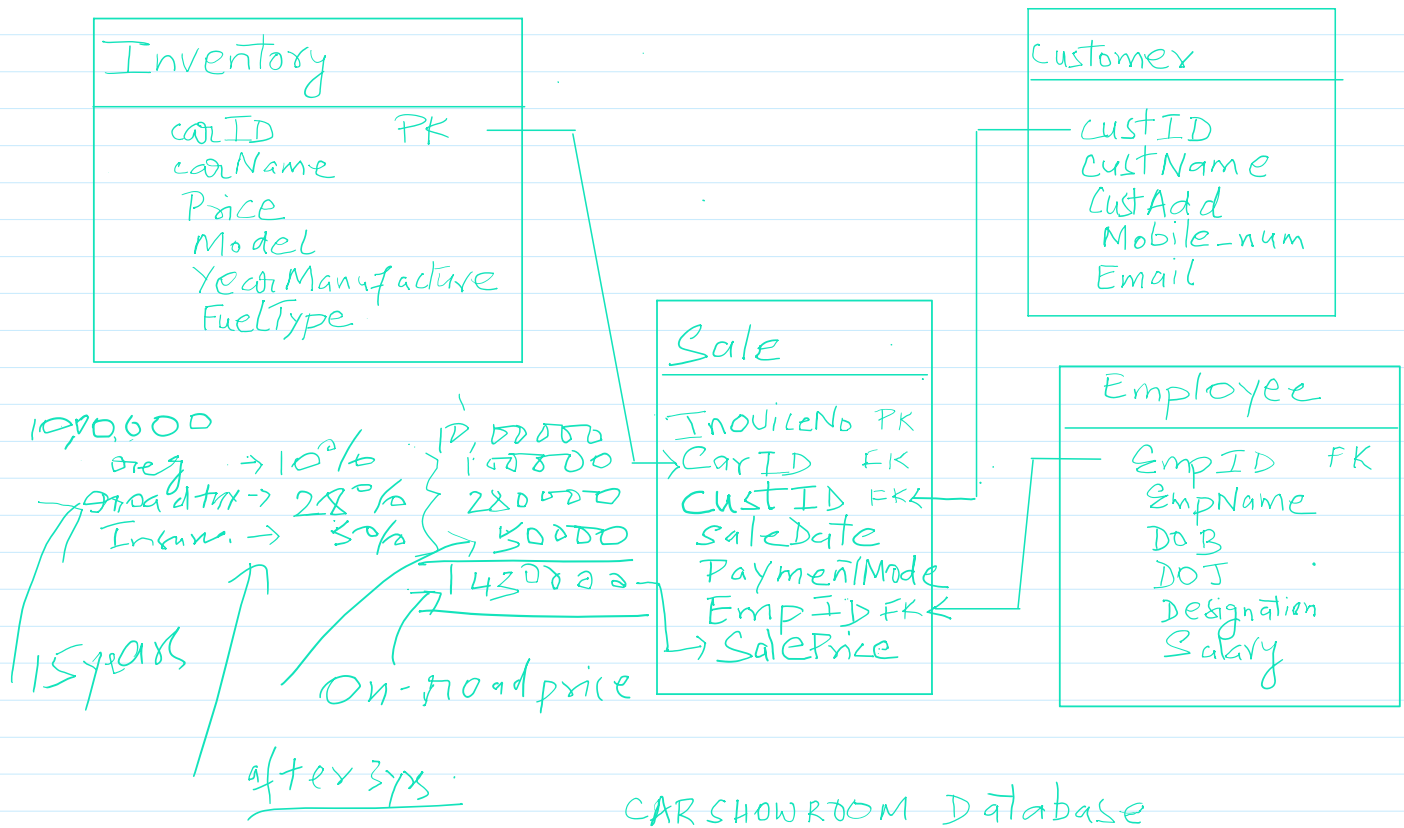


Functions in SQL

08 September 2026 09:24

We will understand how to use single row function, multiple row function, group records based on some criteria, and working on multiple tables using SQL.

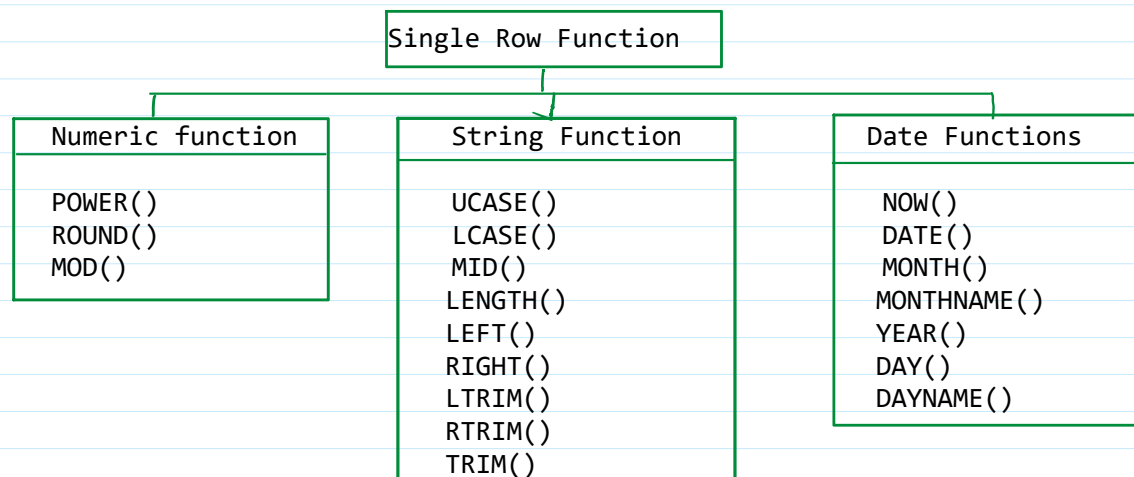


CAR SHOWROOM Database

- * Create database.
- * Create relations (tables)
- * insert values in the relations (tables).

Single Row Functions

These are also known as Scaler Functions. Single row functions are applied on a single value and return a single value. Different single row functions under three categories - numeric (math) String, Date and time.



LTRIM()
RTRIM()
TRIM()

DAYNAME()

Math function except numeric value as input and return a numeric value as a result. String function accept character value as the input and return either character or numeric values as output. Date and time function except date and time value as input and return numeric or string or date and time as output.

Math Functions

Function	Description	Example with output
POWER(X, Y)	Calculates X to the power Y.	<pre>mysql> SELECT POWER(2, 3); +-----+ POWER(2, 3) +-----+ 8 +-----+ 1 row in set (0.144 sec) mysql> SELECT POWER(4, 0.5); +-----+ POWER(4, 0.5) +-----+ 2 +-----+ 1 row in set (0.021 sec) mysql> SELECT POWER(9, 0.5); +-----+ POWER(9, 0.5) +-----+ 3 +-----+ 1 row in set (0.032 sec) mysql> SELECT POWER(8, 1/3); +-----+ POWER(8, 1/3) +-----+ 1.999861375368307 +-----+ 1 row in set (0.022 sec) mysql> SELECT POWER(8, 0.33); +-----+ POWER(8, 0.33) +-----+ 1.9861849908740719 +-----+ 1 row in set (0.137 sec) mysql> SELECT POWER(8, 0.3); +-----+ POWER(8, 0.3) +-----+</pre>

		<pre> 1.8660659830736148 +-----+ 1 row in set (0.005 sec) mysql> SELECT POWER(8, 0.34); +-----+ POWER(8, 0.34) +-----+ 2.027918959580058 +-----+ 1 row in set (0.007 sec) mysql> SELECT SQRT(16) -> ; +-----+ SQRT(16) +-----+ 4 +-----+ 1 row in set (0.029 sec) </pre>
ROUND(N, D)	Rounds of number N to D number of decimal places. Note: If D equal to zero then it round off the number to the nearest integer.	<pre> mysql> SELECT ROUND(283.2); +-----+ ROUND(283.2) +-----+ 283 +-----+ 1 row in set (0.142 sec) mysql> SELECT ROUND(283.5); +-----+ ROUND(283.5) +-----+ 284 +-----+ 1 row in set (0.009 sec) mysql> SELECT ROUND(283.5671, 2); +-----+ ROUND(283.5671, 2) +-----+ 283.57 +-----+ 1 row in set (0.005 sec) </pre>
MOD(A, B)	Return the remainder after dividing number A by number B.	<pre> mysql> SELECT MOD(25, 7); +-----+ MOD(25, 7) +-----+ 4 +-----+ 1 row in set (0.023 sec) </pre>

Homework:

1. Calculate GST as 18% of price and display the result after rounding it off to one decimal place.

2. Add a new column FinalPrice to the table inventory which will have the values as sum of price and 18% of GST.
3. Calculate and display the amount to be paid each month (in multiples of 1000) which is to be calculated after dividing the final price of the car into ten installments.
4. After dividing the amount into EMI's find out the remaining amount to be paid immediately by performing modular division.
5. Let us now add a new column Commission to the sales table. The column Commission should have total length of 7 in which two decimal places to be there.
6. Let us now calculate Commission for sales agents as 12% of the sale price. Insert the values to the newly added column Commission and then display records of the sales where Commission is greater than 73,000.
7. Display invoice number, sale price and commissions such that Commission value is rounded off to 0.

String Functions

String functions can perform various operations on alphanumeric data which are stored in a table. They can be used to change. The case (upper case to lower case or vice versa) extract a substring, Calculate the length of a string and so on.

Table of String functions

Function	Description	Example with Output
UCASE(string) OR UPPER(string)	Convert string into uppercase.	<pre>mysql> select UCASE("lisha"); +-----+ UCASE("lisha") +-----+ LISHA +-----+ 1 row in set (0.050 sec)</pre> <pre>mysql> select UPPER("lisha"); +-----+ UPPER("lisha") +-----+ LISHA +-----+ 1 row in set (0.014 sec)</pre>
LOWER(string) or LCASE(string)	Convert string into lower case.	<pre>mysql> select LOWER("LISHA"); +-----+ LOWER("LISHA") +-----+ lisha +-----+ 1 row in set (0.053 sec)</pre> <pre>mysql> select LCASE("LISHA"); +-----+ LCASE("LISHA") +-----+ lisha +-----+ 1 row in set (0.021 sec)</pre>
MID(string, pos, n) or SUBSTRING(string, pos, n) or SUBSTR(string, pos, n)	Returns a substring of size n starting from the specified position (pos) of	<pre>mysql> SELECT MID("Informatics", 3, 4); +-----+ MID("Informatics", 3, 4) +-----+ form </pre>

	the string. If n is not specified, it returns the substring from the position till end of the string.	<pre> +-----+ 1 row in set (0.022 sec) mysql> SELECT MID("Informatics", 3); +-----+ MID("Informatics", 3) +-----+ formatics +-----+ 1 row in set (0.009 sec) </pre>
LENGTH(string)	Returns the number of characters in the specified string.	<pre> mysql> SELECT LENGTH("Informatics"); +-----+ LENGTH("Informatics") +-----+ 11 +-----+ 1 row in set (0.014 sec) </pre>
LEFT(string, N)	Returns in number of characters from the left side of the string.	<pre> mysql> SELECT LEFT("Infomatics", 4); +-----+ LEFT("Infomatics", 4) +-----+ Info +-----+ 1 row in set (0.065 sec) </pre>
RIGHT(string, N)	Returns N number of characters from the right side of the string.	<pre> mysql> SELECT RIGHT("Informatics", 4); +-----+ RIGHT("Informatics", 4) +-----+ tics +-----+ 1 row in set (0.133 sec) </pre>
INSTR(string, substring)	Returns the position of the first occurrence of the substring in the given string. Returns zero if the substring is not present in the string.	<pre> mysql> SELECT INSTR("Informatics", "ma"); +-----+ INSTR("Informatics", "ma") +-----+ 6 +-----+ 1 row in set (0.042 sec) </pre>
LTRIM(string)	Returns the given string after removing leading white space characters.	<pre> mysql> SELECT LTRIM(" Informatics"); +-----+ LTRIM(" Informatics") +-----+ Informatics +-----+ 1 row in set (0.046 sec) mysql> SELECT LENGTH(" Informatics"); +-----+ LENGTH(" Informatics") +-----+ 19 +-----+ 1 row in set (0.009 sec) mysql> SELECT LENGTH(LTRIM(" </pre>

		<pre> Informatics")); +-----+ LENGTH(LTRIM(" Informatics")) +-----+ 11 +-----+ 1 row in set (0.009 sec) </pre>
RTRIM(string)	Returns the given string after removing trailing white space characters.	<pre> mysql> SELECT LENGTH("Informatics "); +-----+ LENGTH("Informatics ") +-----+ 20 +-----+ 1 row in set (0.029 sec) mysql> SELECT LENGTH(RTRIM("Informatics ")); +-----+ LENGTH(RTRIM("Informatics ")) +-----+ 11 +-----+ 1 row in set (0.027 sec) </pre>
TRIM(string)	Returns the given string after removing both leading and trailing white space characters.	<pre> mysql> SELECT LENGTH(" Informatics "); +-----+ LENGTH(" Informatics ") +-----+ 29 +-----+ 1 row in set (0.013 sec) mysql> SELECT LENGTH(TRIM(" Informatics ")); +-----+ LENGTH(TRIM(" Informatics ")) +-----+ 11 +-----+ 1 row in set (0.051 sec) </pre>

Display customer name in lowercase and customer email in uppercase from table customer.

```

mysql> SELECT LOWER(custname), UPPER(email) from customer;
+-----+-----+
| LOWER(custname) | UPPER(email) |
+-----+-----+
| qasim chauhan   | QASIM.CHAUHAN@GMAIL.COM |
| ashok chowdhury | ASHOK.CHOWDHURY@GMAIL.COM |
| vikas saxena   | VIKAS.SAXENA@GMAIL.COM |
| naveen sen     | NAVEEN.SEN@GMAIL.COM |
| harish malhotra | HARISH.MALHOTRA@GMAIL.COM |
| aditya saxena  | ADITYA.SAXENA@GMAIL.COM |
| vikas pandey   | VIKAS.PANDEY@GMAIL.COM |
+-----+-----+

```

Display the length of the email and the part of the email from the email ID before the character '@'. Note do not print '@'.

```
mysql> SELECT LENGTH(email), LEFT(email, INSTR(email,"@")-1) FROM customer;
+-----+-----+
| LENGTH(email) | LEFT(email, INSTR(email,"@")-1) |
+-----+-----+
| 21 | aarav.joshi |
| 22 | aarav.pillai |
| 23 | aarav.rathore |
| 22 | aarav.saxena |
| 20 | aarti.bose |
| 26 | abhishek.agarwal |
| 25 | abhishek.bakshi |
| 25 | abhishek.kapoor |
+-----+-----+
```

Let us assume that four - digit area code is reflected in the mobile number starting from the position #3. For example, 1851 is the area code of the mobile number 9818511338. Now write the SQL query to display custID, custName, the area code of the customer living in 'Guwahati'.

```
mysql> SELECT custID, CustName, MID(MobileNum, 3, 4) FROM CUSTOMER WHERE CustAddress like '%Guwahati%';
+-----+-----+-----+
| custID | CustName | MID(MobileNum, 3, 4) |
+-----+-----+-----+
| 76 | Madhavi Chowdhury | 0671 |
| 96 | Mamta Joshi | 7320 |
| 144 | Aarav Joshi | 7995 |
| 146 | Nikhil Dutta | 9915 |
| 198 | Kavya Bhatia | 5244 |
+-----+-----+-----+
5 rows in set (0.025 sec)
```

After removing the domain name extension like '.com' from emails of customers.

```
mysql> SELECT TRIM(".com" FROM email) FROM customer;
+-----+
| TRIM(".com" FROM email) |
+-----+
| aarav.joshi@gmail |
| aarav.pillai@gmail |
| aarav.rathore@gmail |
| aarav.saxena@gmail |
| aarti.bose@gmail |
| abhishek.agarwal@gmail |
| abhishek.bakshi@gmail |
| abhishek.kapoor@gmail |
| abhishek.sharma@gmail |
| abhishek.sharmal@gmail |
| aditya.menon@gmail |
| aditya.nayak@gmail |
| aditya.rathore@gmail |
| aditya.saxena@gmail |
| ajay.chowdhury@gmail |
+-----+
```

Homework

Using the table employee from car showroom database write SQL queries for the following:

1. Display employee name and the last two characters of his employee ID.
2. Display designation of employee and the position of character 'E' in the designation if present.
3. Display the details of all the customers having yahoo emails only.